

Name _____

Exam III-A
Make-Up

Score _____

Section I. Short Answer Problems. Answer the problems on separate paper. You do not need to rewrite the problem statements on your answer sheets. Work carefully. Do your own work. **Show all relevant supporting steps!** *STAPLE* this sheet to the front of your answers.

1. (4 pts) Consider the vector space $\mathbb{R}^4$ with basis
 $B = \{(1, 3, 0, -2)^T, (-1, 2, 0, -3)^T, (2, -1, 1, 0)^T, (0, -1, 2, 1)^T\}$. Suppose that
 $[x]_B = (3, 1, -4, -2)^T$. Find the standard coordinates of x .

2. (8 pts) Consider the vector space $\mathbb{R}^4$ with basis
 $B = \{(1, -1, 1, 0)^T, (0, -1, 1, -1)^T, (1, 0, -1, 1)^T, (-1, 1, 0, -1)^T\}$. Suppose that
 $x = (3, 1, -4, -2)^T$. Find $[x]_B$.

3. (10.5 pts) Consider the vector space $\mathbb{R}^5$. Consider the vectors
 $u = (-2, -2, 1, -3, 1)^T$, $v = (1, -2, 1, 1, 0)^T$, $w = (3, 0, -1, -2, -2)^T$. Find

$$A. \|w\|, B. \|u+v\|, C. d(v, w), D. \cos \theta,$$

where θ is the angle between u and v . Also, determine whether any of the following pairs of vectors are orthogonal, parallel or neither:

- a. u and v , b. v and w , c. $v-w$ and u

4. (10.5 pts) Consider the inner product space $\mathbb{R}^5$ with inner product
 $\langle u, v \rangle = 4u_1v_1 + 3u_2v_2 + 2u_3v_3 + 2u_4v_4 + u_5v_5$ for
 $u = (u_1, u_2, u_3, u_4, u_5)^T$ and $v = (v_1, v_2, v_3, v_4, v_5)^T$. Consider the vectors
 $u = (-2, -2, 1, -3, 1)^T$, $v = (1, -2, 1, 1, 0)^T$, $w = (3, 0, -1, -2, -2)^T$. Find

$$A. \|w\|, B. \|u+v\|, C. d(v, w), D. \cos \theta,$$

where θ is the angle between u and v . Also, determine whether any of the following pairs of vectors are orthogonal, parallel or neither:

- a. u and v , b. v and w , c. $v-w$ and u

5. (10.5 pts) Consider the inner product space P_2 with (coefficient) inner product $\langle p, q \rangle = 2a_0b_0 + a_1b_1 + a_2b_2$ for $p(x) = a_0 + a_1x + a_2x^2$ and $q(x) = b_0 + b_1x + b_2x^2$. Consider the vectors $u = 1 + 2x - x^2$, $v = -x - 4x^2$, $w = 3 + 2x - 2x^2$. Find

$$A. \|w\|, B. \|u+v\|, C. d(v, w), D. \cos \theta,$$

where θ is the angle between u and v . Also, determine whether any of the following pairs of vectors are orthogonal, parallel or neither:

- a. u and v , b. v and w , c. $v-w$ and u

6. (8 pts) Consider the inner product space $\mathbb{R}^5$ with inner product $\langle u, v \rangle = 4u_1v_1 + 3u_2v_2 + 2u_3v_3 + 2u_4v_4 + u_5v_5$ for $u = (u_1, u_2, u_3, u_4, u_5)^T$ and $v = (v_1, v_2, v_3, v_4, v_5)^T$. Consider the vectors $u = (-2, -2, 1, -3, 1)^T$, $v = (1, -2, 1, 1, 0)^T$, $w = (3, 0, -1, -2, -2)^T$. Find $\text{proj}_v u$ and $\text{proj}_w v$.

7. (10 pts) Consider the inner product space $\mathbb{R}^3$ with standard inner product (dot product). Let S be the subspace of $\mathbb{R}^3$ which is the span of the vectors $u = (0, 1, 1)^T$ and $v = (-1, 0, 1)^T$. Find the projection of $w = (-4, 1, 2)^T$ on to S .